

Subject 15: Comparing CLOSEST occupation-density basins with direct trajectory support

Definitions, reconstructed evidence, and implications for elementary occupation types

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Abstract

This report examines whether maximum basins reconstructed by discrete CLOSEST ascent should be identified directly with elementary occupation types (EOTs). It uses the selected Subject 15 occupation-density field, its 6,529-vertex graph, the 352 reconstructed maximum basins, and all 70 observed subject visits. The analysis separates three notions that can otherwise be conflated: the number of graph vertices assigned to a basin, the occupation-density probability mass assigned to it, and the number of observed subject trajectory vertices that fall in it. All observed visits fall in M1–M17, and those 17 basins carry numerically all density mass. Density mass and direct visit frequency agree closely, whereas graph support is only moderately associated with direct visit frequency. The singleton M17 basin is a valid discrete local maximum: its zero-step ascent path terminates at itself even though each neighboring vertex chooses a closer uphill route to M1. These results support treating CLOSEST basins as candidate geometric catchments and applying an explicit subject-support criterion before interpreting them as major EOTs.

Key finding. For Subject 15, the apparent proliferation to 352 local-maximum basins is almost entirely a numerical tail phenomenon: M1–M17 contain all 70 observed visits and more than 99.999999999% of the reconstructed density mass. Direct trajectory support, not graph support alone, is therefore the most transparent additional criterion for deciding which candidate basins are scientifically material.

1 Question and data

The scientific question is not whether the discrete CLOSEST construction is internally valid. It is whether every basin produced by that construction should be interpreted as a major elementary occupation type. Smooth-field intuition suggests that a local maximum has a nontrivial attraction neighborhood. On a finite, nonsmooth graph, however, a strict local maximum can have a basin containing only itself: every adjacent vertex may have another higher neighbor connected by a shorter edge.

The analysis uses the selected field HMP_S15_K03_ETA_04 with heat-kernel scale $\eta = 0.953$. The graph has 6,529 vertices and is connected. Subject 15 contributes 70 observed vertices at time indices 1–72, with indices 43 and 65 unobserved. The continuous midpoint-exposure span is 71 days. Basin labels M1, M2, ... are ordered by descending trajectory-flow density mass, matching the default labeling rule used in the application.

2 Definitions

2.1 Graph, field, and closest ascent

Let $G = (V, E)$ be the weighted undirected graph, with edge length $\ell(v, u) > 0$ for $\{v, u\} \in E$. The selected normalized occupation-density field is

$$\hat{\rho} : V \rightarrow [0, 1], \quad \sum_{v \in V} \hat{\rho}(v) = 1.$$

A vertex m is a strict discrete local maximum if $\hat{\rho}(m) > \hat{\rho}(u)$ for every neighbor u of m . The implemented construction also handles exact plateaus canonically; no relevant maximum in the present M17 example is a plateau.

For a nonmaximum vertex v , define its CLOSEST ascending successor by

$$T(v) = \arg \min_{u \sim v, \hat{\rho}(u) > \hat{\rho}(v)} \ell(v, u),$$

with the implementation's deterministic tie handling. A local maximum has no higher neighbor and is a root. Repeated application of T defines

$$r(v) = T^{(k)}(v) = m$$

at the first root m . A maximum therefore belongs to its own basin by the zero-step path $r(m) = m$; the definition does *not* require another vertex to flow to it.

2.2 Basin characteristics

For a local maximum m , its CLOSEST basin is

$$B_m = \{v \in V : r(v) = m\}.$$

The quantities compared in this report are:

Graph support

$S_m = |B_m|$, the number of all graph vertices whose discrete ascent root is m .

Density mass

$P_m = \sum_{v \in B_m} \hat{\rho}(v)$, the occupation-density probability mass allocated to the basin.

Peak value

$H_m = \hat{\rho}(m)$.

Merge level and prominence

If the branch born at H_m dies at merge level D_m under the selected continuation rule, its prominence is $\Pi_m = H_m - D_m$. A component survivor is handled separately by the merge-tree convention.

Graph support and density mass are properties of the field on the complete graph. They are not counts of observed subject visits.

2.3 Direct trajectory support and temporal summaries

Let $x_1, \dots, x_n$ be the observed Subject 15 graph vertices in temporal order, with $n = 70$. Direct visit support for basin m is

$$N_m = \sum_{i=1}^n \mathbf{1}\{r(x_i) = m\}, \quad F_m = \frac{N_m}{n}.$$

An observed run is a maximal consecutive block in the observed sequence with the same basin label. The report records the number of runs, the longest run, and the number of returns (runs minus one for a visited basin).

For descriptive duration sensitivity, observation i receives a midpoint (one-dimensional Voronoi) weight:

$$w_1 = \frac{t_2 - t_1}{2}, \quad w_n = \frac{t_n - t_{n-1}}{2}, \quad w_i = \frac{t_{i+1} - t_{i-1}}{2} \quad (1 < i < n).$$

The basin exposure is $W_m = \sum_i w_i \mathbf{1}\{r(x_i) = m\}$ and its share is $W_m / \sum_i w_i$. These weights summarize represented observation time over the observed interval. They are *not* an EOT estimator and do not by themselves establish quasi-equilibrium behavior.

2.4 Agreement and ranking

All ranks are descending, with tied values receiving the minimum rank. Across M1–M17, monotone agreement is summarized by Spearman rank correlation. Distributional discrepancy between density mass P and visit share F is summarized by total variation distance,

$$d_{\text{TV}}(P, F) = \frac{1}{2} \sum_m |P_m - F_m|.$$

3 Reconstruction and validation

The report is generated from the archived Subject 15 scale-path basin asset, the exact graph, the visit metadata, and the committed Subject 15 merge-tree fixture. The derivation script independently verifies the following conservation and identity conditions before writing any result:

- all 6,529 graph vertices have one assigned maximum root;
- the reconstructed basin supports sum to 6,529;
- basin density masses sum to one within numerical tolerance;
- direct basin counts sum to 70 visits and midpoint weights sum to 71 days;
- current and archived reconstructions agree for every root and successor vertex;
- exactly M1–M17 have direct Subject 15 visits.

The full 352-row comparison table and the 70-row visit assignment table are retained as machine-readable CSV files rather than compressed into the PDF.

4 Results

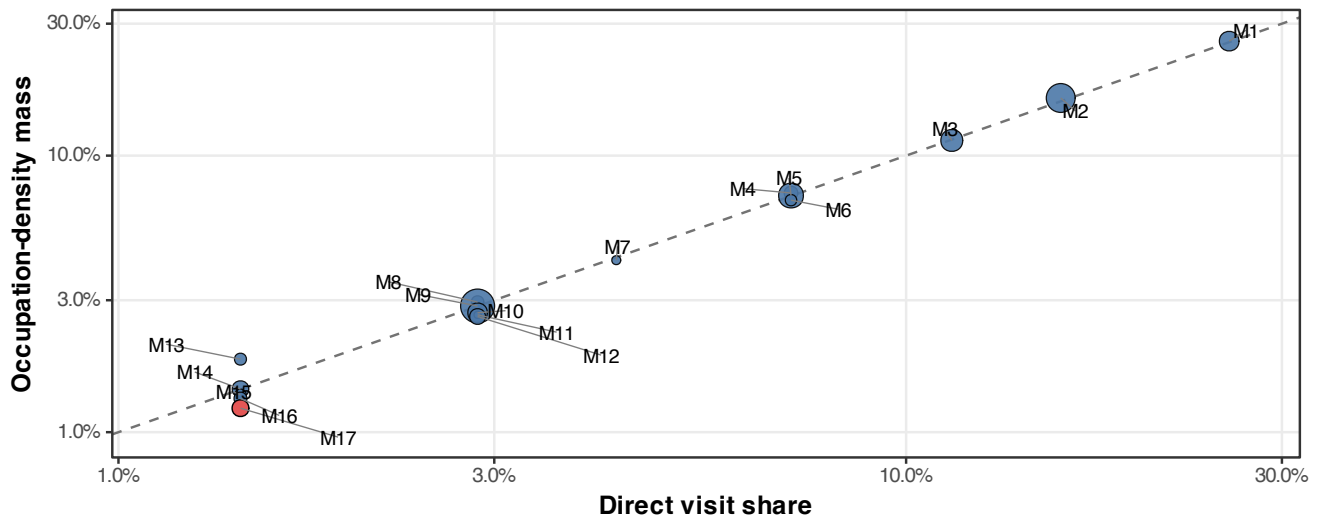
4.1 Density mass closely tracks direct visit frequency

Only 17 of the 352 maximum basins are directly visited. They are exactly M1–M17 and together carry more than 99.9999999999% of the density mass. The remaining 335 basins carry only approximately 8.7×10^{-14} mass in total. Among M1–M17, density mass and direct visit count have Spearman correlation $\rho = 0.973$. Across all 352 basins, the total variation distance between density mass and empirical visit share is only 1.39%.

4.2 The 17 directly observed basins

A. Density mass tracks direct visit frequency

Dashed line is exact agreement; both axes are logarithmic.



B. Basin-level discrepancies are small

Positive values mean density mass exceeds direct visit share.

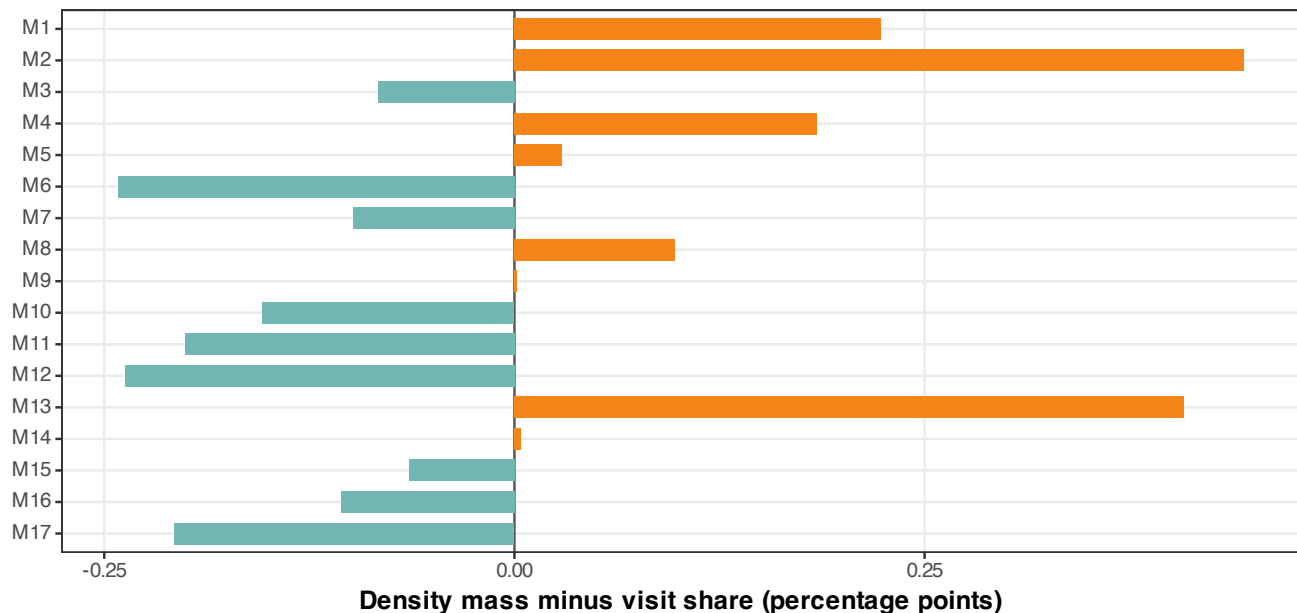


Figure 1: Comparison of reconstructed occupation-density mass with direct Subject 15 visit frequency. M17 is highlighted in red. The density estimate and raw visit distribution agree closely, despite being conceptually distinct summaries.

Table 1: Characteristics of all maximum basins with at least one direct Subject 15 visit. “Mass” is percentage occupation-density mass; “Days” is midpoint exposure and is descriptive only.

Basin	Peak vertex	Graph support	Visits	Runs	Longest run	Mass (%)	Days
M1	1,598	316	18	8	5	25.94	18.0
M2	1,628	715	11	5	7	16.16	11.0
M3	1,635	399	8	2	7	11.35	8.5
M4	1,575	141	5	4	2	7.33	5.5
M5	1,641	517	5	3	3	7.17	5.0
M6	1,578	95	5	3	3	6.90	5.0
M7	1,609	55	3	3	1	4.19	3.0
M8	1,603	118	2	2	1	2.95	2.0
M9	1,622	1,003	2	2	1	2.86	2.5
M10	1,590	312	2	1	2	2.70	2.0
M11	1,614	32	2	1	2	2.66	2.0
M12	1,621	197	2	1	2	2.62	2.0
M13	1,638	107	1	1	1	1.84	1.0
M14	1,574	217	1	1	1	1.43	0.5
M15	1,618	98	1	1	1	1.36	1.0
M16	1,640	126	1	1	1	1.32	1.0
M17	1,589	1	1	1	1	1.22	1.0

M1 is both the most massive and most frequently visited basin. The table also shows why no single support quantity is sufficient: M9 contains 1,003 graph vertices but only two direct visits, whereas M17 contains one graph vertex and one direct visit.

4.3 Graph support is not empirical trajectory support

Across M1–M17, graph support and direct visit count have Spearman correlation $\rho = 0.464$, much weaker than the mass–visit association. Graph support measures the size of a geometric attraction set over every vertex in the smoothed ambient graph. Direct visit count measures how many observed subject states land in that set. The two may diverge because the graph contains interpolating or other-subject states, sampling is sparse, and CLOSEST routing is determined by local edge lengths.

Graph support and direct trajectory support are different quantities

M17 is a one-vertex graph basin and one observed visit; Spearman $\rho = 0.464$ among M1–M17.

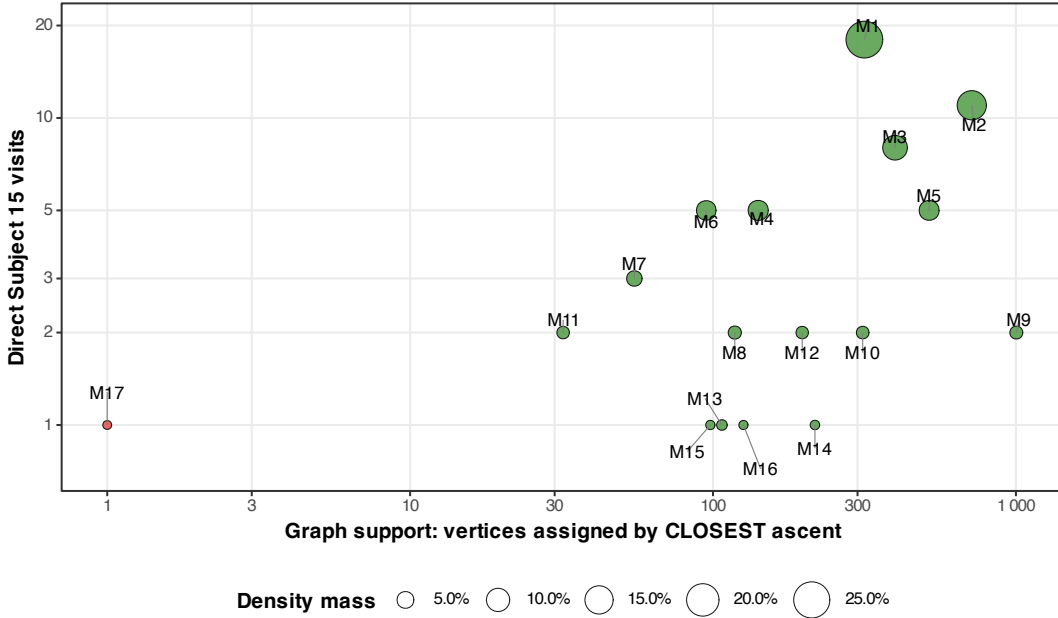


Figure 2: Graph support versus direct Subject 15 visit count for M1–M17. Point area represents density mass. M17 is highlighted in red.

4.4 Why M17 can have support one

M17 is the strict local maximum at vertex 1589. Its peak and density mass are both 0.0122134; it is mass rank 17, graph-support rank 304, and has prominence 0.00196728. It merges into M1 at field level 0.0102461. The only direct Subject 15 visit is time index 16 (UAB015_W3D2, state S3, CST IV-B), representing one midpoint day.

The apparent paradox disappears when the local routing definition is applied. All three neighbors lie below M17, so vertex 1589 is a local maximum and roots at itself. Yet each neighbor has a different higher vertex connected by a shorter edge than its edge to M17. Consequently, none chooses M17 as its CLOSEST ascending successor.

Table 2: Exact local CLOSEST routing around M17 (vertex 1589). “Chosen length” is the edge to the selected higher successor; “M17 length” is the edge from that source vertex to M17.

Source	Field	Chosen successor	Chosen length	M17 length	Final root
1,588	0.008493	1,584	0.07211	0.10430	M1
1,592	0.010246	1,597	0.08153	0.08246	M1
1,593	0.010246	1,592	0.05646	0.11791	M1

Thus support one is not an error: $B_{M17} = \{1589\}$. It is evidence that smooth catchment intuition does not transfer automatically to a nonsmooth finite graph. M17 remains empirically relevant because the subject actually visits its root once; many numerical-tail singleton basins do not.

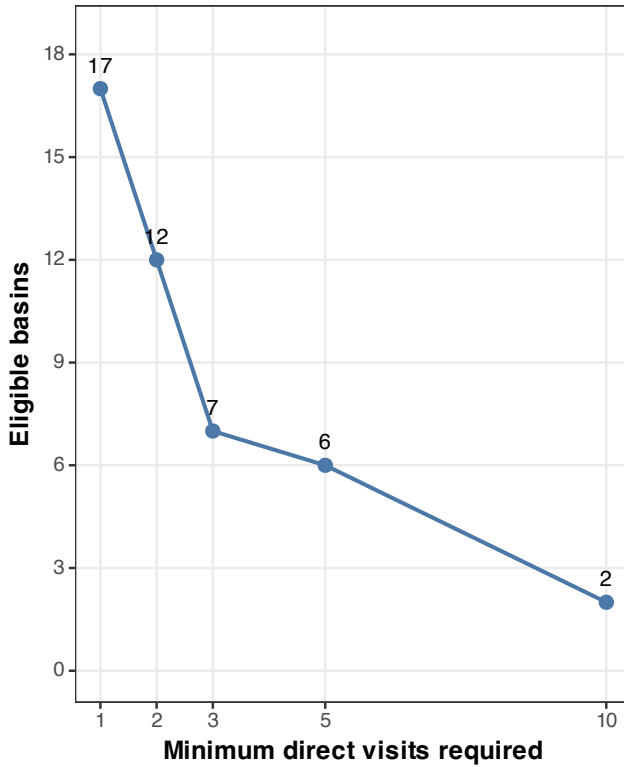
4.5 Visit-threshold sensitivity

A direct-support threshold is scientifically interpretable but it changes the estimand. The following is a sensitivity analysis, not a recommended final cutoff.

Table 3: Consequences of requiring a minimum number of direct Subject 15 visits. Coverage columns are percentages.

Min. visits	Basins	Visits	Visit coverage	Days	Time coverage	Mass coverage
1	17	70	100.0	71.0	100.0	100.0
2	12	65	92.9	66.5	93.7	92.8
3	7	55	78.6	56.0	78.9	79.0
5	6	52	74.3	53.0	74.6	74.8
10	2	29	41.4	29.0	40.8	42.1

A. Number of retained candidate basins



B. Coverage retained under each threshold

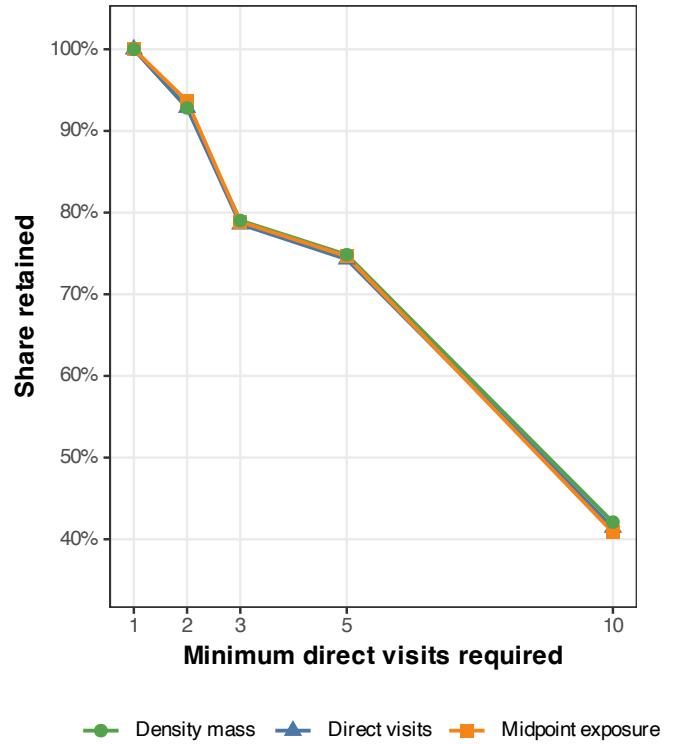


Figure 3: Sensitivity of the candidate EOT set and retained coverage to a direct-visit threshold. Visit, midpoint-exposure, and density-mass coverage remain similar for this subject.

Requiring at least two visits retains 12 basins and about 93% of visits, midpoint exposure, and density mass. Requiring at least three retains seven basins and about 79% of each. A hard threshold therefore provides interpretability at the cost of discarding genuine low-frequency states such as M17. A more defensible final rule may use a primary visit threshold plus an explicit rare-state category or stability criterion.

4.6 Temporal organization

Counts alone do not distinguish one sustained episode from repeated returns. The observed sequence shows both long occupancy blocks and isolated visits. M17 is a single observed state, while M1 occurs in eight runs and M2 in five. This motivates reporting run count and persistence alongside total visits.

Observed Subject 15 basin sequence

Seventy observed visits over time indices 1–72; missing indices 43 and 65 create visible gaps.

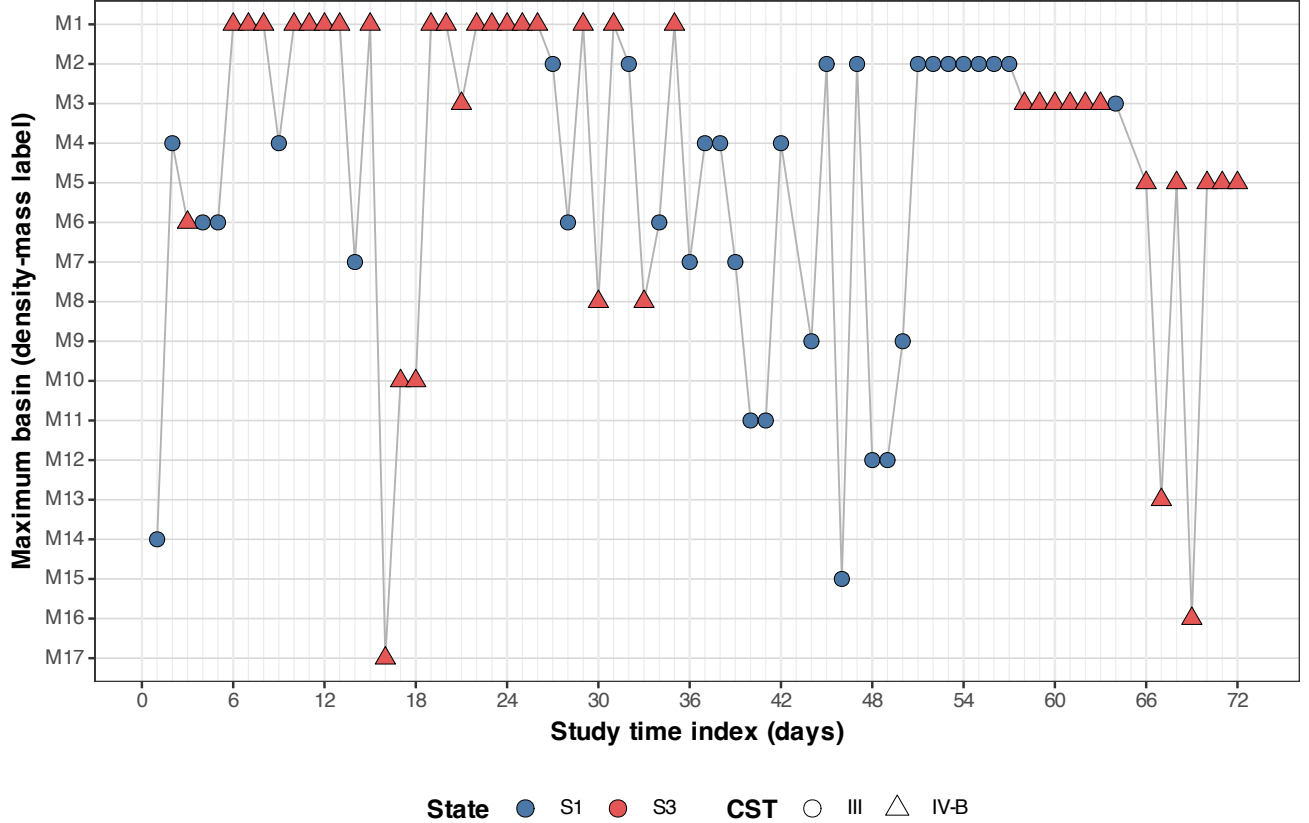


Figure 4: Temporal sequence of directly observed basin assignments. Horizontal position is study time index; vertical position is the density-mass basin label. Color and shape retain the observed state and CST annotations.

5 Implications for elementary occupation types

The evidence supports a two-stage interpretation:

1. **Candidate construction.** Use the canonical CLOSEST reconstruction to partition the occupation-density field into geometrically well-defined attraction basins.
2. **Scientific qualification.** Decide which candidates count as major EOTs using subject-facing evidence: direct visit count or share, temporal persistence and returns, density mass, and stability across smoothing or graph scales.

This separation preserves the mathematical value of the canonical basin complex without asserting that every discrete local maximum is an elementary occupation type. It also makes the choice of estimand explicit:

- **Visit-weighted EOT prevalence** treats each observed state equally and answers how frequently

sampled states fall in each basin.

- **Duration-weighted occupation** uses represented time and moves toward a time-occupation or quasi-equilibrium target. It is useful as sensitivity analysis but should not silently replace an EOT frequency estimand.
- **Graph support** describes the ambient reconstructed field and should not be interpreted as subject trajectory support.

For this data set, raw visit share and midpoint-weighted share are close, so the practical ranking is robust to the simple duration weighting. That agreement is an empirical result for Subject 15, not a general equivalence.

6 Recommended next analysis

Before fixing a universal EOT rule, the next comparison should be repeated across subjects and smoothing scales. For each candidate basin, record:

1. direct visits, visit share, runs, longest run, and returns;
2. midpoint or interval exposure as a separate duration-sensitive summary;
3. density mass, peak value, prominence, and graph support;
4. survival and label stability across graph and field scales.

Candidate decision rules can then be compared by retained density mass, retained direct visits, temporal reproducibility, and cross-scale stability. The threshold table in this report is deliberately illustrative; it does not select a final biological cutoff.

7 Limitations

This is a single-subject case study. The graph and occupation-density field depend on upstream embedding, graph construction, connectivity repair, smoothing, and normalization choices. Midpoint weights interpolate represented time but do not model unobserved transitions. Direct visit counts are also sample-design dependent. Finally, no external biological validation is used here; the conclusions concern the relationship among reconstruction geometry, density mass, and observed trajectory support.

8 Reproducible artifacts

The report is rebuilt by `dev/subject15_eot_comparison_2026-08-06/latex_report/build_subject15_eot_comparison_report.sh`. Its figure builder consumes only regenerated CSV/RDS analysis outputs. The principal machine-readable artifacts are:

- `subject15_maximum_basin_eot_comparison.csv`: all 352 maximum basins and all reported characteristics;
- `subject15_visit_basin_assignments.csv`: all 70 observed visits;
- `subject15_eot_visit_threshold_sensitivity.csv`: threshold sensitivity;
- `subject15_m17_local_flow.csv`: exact local M17 routing evidence.

All artifact paths are relative to `dev/subject15_eot_comparison_2026-08-06/`.